

MA 214 Week 14: Lab 5

Bayesian Linear Regression

1. **Your name:** _____

BU ID: _____

2. **Partner's name:** _____

BU ID: _____

Big idea: Bayesian linear regression combines a regression likelihood with prior distributions, then uses posterior draws to describe uncertainty and make predictions. **Do not use GenAI tools for this lab.** The point is to practice your own analysis work, coding skills, and statistical reasoning.

Files used: `Coffee_df.csv` and `lab-starter.R`

Skills practiced: log-posterior functions, `dnorm()`, `dlnorm()`, Metropolis-Hastings output, credible intervals, and posterior prediction.

Your mission: Use Bayesian linear regression to estimate how daily coffee-shop revenue changes with the number of customers and predict revenue for a new day.

Icebreaker

Before opening R: introduce yourself to your partner and answer the warm-up question below.

Warm-up response: If a coffee shop gets one more customer, what do you expect to happen to daily revenue? What could make that relationship noisy?

Getting Started

Open `lab-starter.R`. Load the coffee data and make a first plot of revenue versus customers.

```
library(dplyr)
library(ggplot2)
library(readr)

df <- read.csv("Coffee_df.csv")

head(df)
str(df)

ggplot(df, aes(x = customers, y = revenue)) +
  geom_point()
```

HUDDLE UP

Before calculating anything, does the relationship look roughly linear? What would the slope mean in dollars?

Question 1. Explore the Coffee Data

Identify the response, predictor, and model parameters before writing the log-posterior.

```
x <- df$customers
y <- df$revenue
```

Variable or parameter	Type or role	What it means
revenue		
customers		
beta1		

Write 2-3 sentences describing the relationship before fitting the Bayesian model.

Question 2. Make a Prediction

Before running MCMC, predict whether the posterior for **beta1** will be centered near 0, positive, or negative.

PREDICTION TIME

Use the scatter plot and the prior $\beta_1 \sim N(2, 0.5^2)$ to predict what the posterior slope will look like.

```
log_posterior <- function(beta0, beta1, sigma2) {
  if (sigma2 <= 0) return(-Inf)

  mu <- beta0 + beta1 * x
  log_lik <- sum(dnorm(y, mean = mu, sd = sqrt(sigma2), log = TRUE))

  log_p_beta0 <- dnorm(beta0, mean = 5, sd = 2, log = TRUE)
  log_p_beta1 <- dnorm(beta1, mean = 2, sd = 0.5, log = TRUE)
  log_p_sigma2 <- dlnorm(sigma2, meanlog = 0, sdlog = 0.3, log = TRUE)

  log_lik + log_p_beta0 + log_p_beta1 + log_p_sigma2
}
```

What did your team predict? What evidence did you check first?

Question 3. Compute Posterior Summaries

Run the provided Metropolis-Hastings code in `lab-starter.R`. Then summarize the post-burn-in draws.

```
post_b0 <- beta0_chain[(burn + 1):n_iter]
post_b1 <- beta1_chain[(burn + 1):n_iter]
post_s2 <- sigma2_chain[(burn + 1):n_iter]

beta1_CI <- quantile(post_b1, c(0.025, 0.975))
beta1_CI
```

HUDDLE UP

Does the credible interval for `beta1` support a positive customer-revenue relationship? What does the interval mean in context?

CLAIM, EVIDENCE, CONTEXT

The posterior slope suggests _____.

My evidence is _____.

This matters because _____.

Question 4. Interpret in Context

Use the posterior samples to predict revenue for a new day with $x^* = 3$ customers.

Question	Your answer
What comparison did you make?	
What result is strongest?	
What limitation or caution should readers know?	

DEBATE CORNER

If you and your partner disagree about the prediction, write both arguments first. Then decide what claim the posterior predictive evidence supports.

Question 5. Close the Lab

Submit your completed `lab-starter.R`. Your file should define the required data, log-posterior, credible interval, and posterior predictive mean objects.

Final response: state your posterior predictive claim, cite one posterior summary, and explain the context.

Optional Challenge

Compare posterior predictive draws with and without the extra observation-noise term.

```
# Optional challenge starter code.  
y_mean_only <- post_b0[idx] + post_b1[idx] * x_star  
y_with_noise <- y_mean_only + rnorm(n_samp, 0, sqrt(post_s2[idx]))
```